

**Dr. MANJUNATHA C, Ph.D. MRSC**

Associate Professor, Department of Chemistry,
 Head, Centre of Excellence in Nanomaterials and Devices (CND)
 RV College of Engineering, Bengaluru-560059, INDIA
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<https://orcid.org/0000-0003-0422-9614>
<https://scholar.google.co.in/citations?user=xGChruoAAAAJ&hl=en>

Member of Royal Society of Chemistry (RSC), UK., Electrochemical Society, (ECS) and American Chemical Society (ACS), USA.

TEACHING EXPERIENCE: 23 Years (RVCE 18 Years)

Applied/Engineering Chemistry, Physical Chemistry, Electrochemistry, Inorganic Chemistry
 Technical Chemistry, Chemistry of Smart materials and Devices, Chemistry of Functional Materials,
 Nanomaterials for UG students of all programs of Engineering

RESEARCH EXPERTISE and INTERESTS: 18 Years

- ❖ Nanoscience and Technology, Carbon/Inorganic Nano Materials/composite
- ❖ Electrochemical energy storage (Li ion battery and Supercapacitors),
- ❖ Energy conversion (Green Hydrogen)
- ❖ Sensor (Electrochemical and Gas) applications,
- ❖ Modelling/Simulation of Li battery Digital twinning, Electrochemical modelling, Li ion battery technology for EVs, Cell/Battery testing.

RESEARCH HIGHLIGHTS/ACCOMPLISHMENTS/AWARDS

1. Listed among **WORLD'S TOP 2% SCIENTISTS in 2023**, in the field of Materials Science-Energy, by **Stanford University**-Scopus survey.
2. International **"ICTSGS SERVICE AWARD"** sponsored by "American Chemical Society" (ACS) and "Electrochemical Society" (ECS) in 2021 by **Yamagata University, JAPAN**.
3. RV College of Engineering **"BEST RESEARCHER AWARD"** for the academic year 2019-2020 by Indian Society for Technical Education (ISTE), RVCE chapter, INDIA.
4. **Two times, "EXCELLENT PERFORMER"** Award for the academic year 2021-22 and 2022-23 by **"Rashtreeya Sikshana Samithi Trust" (RSST)**, Bengaluru, INDIA.
5. **"BEST STUDENT PROJECT AWARD"** by Indian Institute of Chemical Engineers (IIChe), Bangalore Chapter, (The Institute is recognized by the Department of Science and Technology, Government of India.), 26th July 2022.
6. **Awarded International Travel Support (ITS)** from SERB-DST, Government of India (Sanction No. ITS/2019/000577) to deliver an invited talk at the World Nanotechnology Conference, held in Dubai, UAE, April 15-17, 2019.
7. **Recognized for publishing three research articles selected as front cover features** in internationally acclaimed journals: *ACS Applied Electronic Materials*, *Energy & Fuels* (ACS), *ChemSusChem* (Wiley – Chemistry Europe)
8. **Journal Publications = 120 articles**, (First Author : 23, Corresponding Author: 98, Q-1 = 42; Q-2 = 27; Q-3 = 11; Q-4 = 03 , Scopus/SCI/WoS = 95, Patents = 06, Citations: 2500, H-index = 30, i10 index = 61, Cumulative-Impact Factor = 430 and Cumulative-Cite score = 645)

ADMINISTRATIVE RESPONSIBILITY: Department: NAAC, NBA, AICTE, VTU, CIE-TEST, TIME TABLE, SCIENCE FORUM, RESEARCH LAB, COUNSELOR **Institution:** CoE Centre for Nanomaterials and Devices, RVJSTEAM journal, NAAC, Research Advisory committee.

FULL NAME	MANJUNATHA C	
Date of Birth	Nationality	Gender
20 July 1981	Indian	Male

EDUCATION

PhD

Degree Award	PhD	Faculty of Science- Applied Chemistry
Name of university	Visvesvaraya Technological University (VTU), Belagavi-590018, INDIA	
Name of the research centre, and supervisor	Department of Chemistry, MS Ramaiah institute of Technology , Bengaluru, Supervisor, Prof BM Nagabhushana	
Thesis Title/ Year	Synthesis, Characterization and Luminescence Studies on Doped and un-doped CdSiO ₃ Nanophosphor	2009-2013
Research Domain/Area	Materials Science-Nanomaterials Synthesis/Characterisation/Electrochemical Energy Storage, Conversion and Sensor Applications	

NET (National Eligibility Test)

Certificate of Qualifying exam	CSIR-UGC-NET (National Eligibility Test) Chemical Sciences
Rank	42 (All India Rank)
Name of Awarding Agency	CSIR-UGC, MHRD, Govt of INDIA
Year of Completion	22-06-2014

PG

Degree Award / Position	M.Sc (Master of Science)	FIRST CLASS
Name of university	Bangalore University, Bengaluru -56001, INDIA	
Title of course / Year	Inorganic Chemistry	2002-2003 to 2003-2004

UG

Degree Award / Position	B.Sc (Bachelors of Science)	FIRST CLASS
Name of university	Bangalore University, Bengaluru-56001, INDIA	
Title of course / Year	Physics, Chemistry, Mathematics	1999-2000 to 2001-2002

PROFESSIONAL EXPERIENCE:**A). Current Role**

Current Job Title	Associate Professor
Dates from/to	Associate Professor 14 December 2024, Till Date
Employer's Name	RV College of Engineering
Employer's Address	Department of Chemistry, RV College of Engineering, Mysuru Road, Bengaluru-560059, INDIA

B). Role prior to A

Current Job Title	Assistant Professor (Selection Grade)	
Dates from/to	Assistant Professor-Selection Grade, August 2024- December 2024. Assistant Professor-Senior Scale, Feb 2023 - August 2024 Assistant Professor, from Feb 2011 to Feb 2023	
Employer's Name	RV College of Engineering	
Employer's Address	Department of Chemistry, RV College of Engineering, Mysuru Road, Bengaluru-560059, INDIA	
Number of PhD , UG, PG Scholars Supervising/Guided	directly	indirectly
	2 PhD (Supervising) 65 UG-PG projects (= 150 Students Guided)	10 UG-PG Students

C) Role prior to B

Job Title	Lecturer
Dates from/to	January 2009 to February 2011
Employer's Name	RV College of Engineering
Employer's Address	Department of Chemistry, RV College of Engineering, Mysore Road, Bengaluru-560059, INDIA

D) Role prior to C

Job Title	Lecturer
Dates from/to	April 2008 to Jan 2009
Employer's Name	Deeksha centre for learning Pre-University College
Employer's Address	Deeksha centre for learning Pre-University College, Kanakapura road, Bengaluru, INDIA
Description of Duties	Teaching Pre-University Chemistry Theory and Practical

E) Role prior to D

Job Title	Lecturer
Dates from/to	June 2005 to April 2008
Employer's Name	The National Pre-University College
Employer's Address	The National Pre-University College, Jayanagar, Bengaluru - 56070, INDIA
Description of Duties	Teaching Pre-University Chemistry Theory and Practical

F) Role prior to E

Job Title	Research Assistant
Dates from/to	Jan 2005 to May 2005
Employer's Name	Indian Institute of Science (IISc)
Employer's Address	Solid State & Structural Chemistry Unit. Indian Institute of Science (IISc), Bengaluru, INDIA
Description of Duties	Research Project Assistant- Inorganic materials synthesis and Charecteristaion

G) Role prior to F

Job Title	Research Project Trainee
Dates from/to	July 2004 to Dec 2004
Employer's Name	General Electric, GE India
Employer's Address	Materials Research Lab. John F Welch Technology Center GEITC. GE Bengaluru, INDIA
Description of Duties	Research Project Assistant- Inorganic materials synthesis and Charecteristaion

PUBLISHED MATERIAL, NUMBER of:

Book Chapters	03	Papers	120	Patents	06
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Patents (Granted/Published/Filed) - 05

1. PATENT APPLICATION PUBLICATION – INDIA-GRANTED	
Title of the invention	Synthesizing NiSe ₂ active electrode nanoparticles for high energy density asymmetric supercapacitor coin cell
Name of the Inventors	1) Manjunatha C , 2) Shwetha K P, 3) Sudha Kamath M K, 4) Yash Athreya, 5) Suraj L, 6) Shweta A Ram, 7) Ananda Islakar, 8) Rajalakshmi M
Name of Applicant	RV College of Engineering
Address of Applicant	Mysore Road, R. V. Vidyaniketan Post, Bangalore – 560059, Karnataka, India. Bangalore
Application Number	202341000518 A
Date of filing / Publication	04/01/2023; 17/02/2023
The Patent Office Journal No./ Page No	07/2023; 11242

2. PATENT APPLICATION PUBLICATION – INDIA-GRANTED	
Title of the invention	A Preparation Method of Tungsten Oxide Nanocubes having High-Specific Capacitance for Electrochemical Energy Storage Applications
Name of the Inventors	1) Manjunatha C , 2) Shwetha K P, 3) Sudha Kamath M K, 4) Yash Athreya, 5) Suraj L, 6) Vinay Kumar, 7) Mamtha V
Name of Applicant	RV College of Engineering
Address of Applicant	Mysore Road, R. V. Vidyaniketan Post, Bangalore – 560059, Karnataka, India. Bangalore
Application Number	202341017602A
Date of filing / Publication	16/03/2023; 14/04/2023
The Patent Office Journal No./ Page No	15/2023; 31703



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GEOGRAPHICAL INDICATIONS



क्रम सं/SL No :044197790



पेटेंट कार्यालय, भारत सरकार

The Patent Office, Government Of India

पेटेंट प्रमाण पत्र

Patent Certificate

(पेटेंट नियमावली का नियम 74)

(Rule 74 of The Patents Rules)

पेटेंट सं. / Patent No.

564019

आवेदन सं. / Application No.

202341000518

फाइल करने की तारीख / Date of Filing

04/01/2023

पेटेंटी / Patentee

R.V. College of Engineering

प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकटित **SYNTHESIZING NiSe₂ ACTIVE ELECTRODE NANOPARTICLES FOR HIGH ENERGY DENSITY ASYMMETRIC SUPERCAPACITOR COIN CELL** नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख जनवरी 2023 के चौथे दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled **SYNTHESIZING NiSe₂ ACTIVE ELECTRODE NANOPARTICLES FOR HIGH ENERGY DENSITY ASYMMETRIC SUPERCAPACITOR COIN CELL** as disclosed in the above mentioned application for the term of 20 years from the 4th day of January 2023 in accordance with the provisions of the Patents Act, 1970.

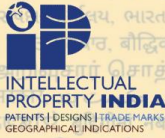


अनुदान की तारीख
Date of Grant : 27/03/2025

पेटेंट नियंत्रक
Controller of Patents

टिप्पणी - इस पेटेंट के नवीकरण के लिए फीस, यदि इसे बनाए रखा जाना है, जनवरी 2025 के चौथे दिन को और उसके पश्चात प्रत्येक वर्ष में उसी दिन देय होगी।

Note. - The fees for renewal of this patent, if it is to be maintained, will fall / has fallen due on 4th day of January 2025 and on the same day in every year thereafter.



क्रम सं/SL No : 044197156



पेटेंट कार्यालय, भारत सरकार

The Patent Office, Government Of India

पेटेंट प्रमाण पत्र

Patent Certificate

(पेटेंट नियमावली का नियम 74)

(Rule 74 of The Patents Rules)

पेटेंट सं. / Patent No.

562332

आवेदन सं. / Application No.

202341017602

फाइल करने की तारीख / Date of Filing

16/03/2023

पेटेंटी / Patentee

R. V. College of Engineering

प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकटित *A Preparation Method of Tungsten Oxide Nanocubes having High-Specific Capacitance for Electrochemical Energy Storage Applications*. नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख मार्च 2023 के सोलहवें दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled *A Preparation Method of Tungsten Oxide Nanocubes having High-Specific Capacitance for Electrochemical Energy Storage Applications*, as disclosed in the above mentioned application for the term of 20 years from the 16th day of March 2023 in accordance with the provisions of the Patents Act, 1970.

अनुदान की तारीख : 11/03/2025
Date of Grant :



पेटेंट नियंत्रक
Controller of Patents

टिप्पणी - इस पेटेंट के नवीकरण के लिए फीस, यदि इसे बनाए रखा जाना है, मार्च 2025 के सोलहवें दिन को और उसके पश्चात प्रत्येक वर्ष में उसी दिन देय होगी।

Note. - The fees for renewal of this patent, if it is to be maintained, will fall / has fallen due on 16th day of March 2025 and on the same day in every year thereafter.

1. PATENT APPLICATION PUBLICATION – INDIA- PUBLISHED	
Title of the invention	A two-step synthesis method for preparing NiSe ₂ @WO ₃ nano-composite
Name of the Inventors	1) Manjunatha C , 2) Shwetha K P, 3) Sudha Kamath M K, 4) Yash Athreya, 5) Suraj L
Name of Applicant	RV College of Engineering
Address of Applicant	Mysore Road, R. V. Vidyaniketan Post, Bengaluru – 560059, Karnataka, India.
Application Number	202341089261
Date of filing of Application	Date/Time 2023/12/28 11:04:42
Date of Publication	04.07.2025
The Patent Office Journal No	27/2025 Dated 04/07/2025 , 65771

2. PATENT APPLICATION PUBLICATION – INDIA- PUBLISHED	
Title of the invention	A Nano Compound for Uric Acid Detection
Name of the Inventors	1) Manjunatha C , 2). Shubha M B, 3). Sudeep Mudhulu, 4). Sham Aan MP, 5) Girish Kumar S
Name of Applicant	RV College of Engineering
Address of Applicant	Mysore Road, R. V. Vidyaniketan Post, Bengaluru – 560059, Karnataka, India.
Application Number	202541057480
Date of filing of Application	16-06-2025
Date of Publication	26-09-2025
The Patent Office Journal No	

3. PATENT APPLICATION PUBLICATION – INDIA- FILED	
Title of the invention	A Scalable Process for Producing Manganese Borate for Asymmetric Supercapacitor Coin Cell Electrodes
Name of the Inventors	1) Manjunatha C , 2) Beena S, 3) Shwetha KP, 4) Chandresh Kumar Rastogi, 5) Sudha Kamath MK
Name of Applicant	RV College of Engineering
Address of Applicant	Mysore Road, R. V. Vidyaniketan Post, Bengaluru – 560059, Karnataka, India.
Application Number	202541065771
Date of filing of Application	Date/Time 2025/07/10; 13:39:32
Date of Publication	26/09/2025
The Patent Office Journal No	39/2025 Dated 26/09/2025, 94626

4. PATENT APPLICATION PUBLICATION – INDIA- FILED	
Title of the invention	Porous Transition-Metal Oxide Nanoparticles for Uric Acid Detection
Name of the Inventors	Manjunatha C, Shubha M B, Sudeep Mudhulu, Shwetha K P, Radha N
Name of Applicant	RV College of Engineering
Address of Applicant	Mysore Road, R. V. Vidyaniketan Post, Bengaluru – 560059, Karnataka, India.
Application Number	202541122447
Date of filing of Application	Date/Time 2025/12/05 12:39:00

Journal paper publications (International peer reviewed Journals) (2012-2026)

2026 Publications	
128	Beena S , Shwetha K.P. , Chandresh Kumar Rastogi , Professor Nelsa Abraham , Suresh Babu Viswanathan , Girish Kumar S , B S Nishchith , Manjunatha C* , Maria Helena Braga*, Borate-enabled redox transport in mixed-metal orthoborate polyanionic frameworks, (reference number: NATSYNTH-26041185-T), submitted to A. 2026
127	Shubha M B, Manjunatha C* , Sudeep Mudhulu, Radha N, Nagendra Babu A P, Bahumanya R G, Shwetha K P, Chandresh Kumar R, 3D Interconnected Porous Spinel NiMn ₂ O ₄ Nanostructures for Interpretable Machine Learning Assisted Uric Acid Sensing, ACS Sensors , Manuscript ID: se-2026-01919k, Submitted, 2026

126	Machine Learning-Guided Design and Electrochemical Performance of Bamboo-Derived Activated Carbon/SrFe ₁₂ O ₁₉ Composite as High-Performance Supercapacitor Electrode , Journal of Materials Science , 2026 (Under review)
125	Nishant Vasantkumar Hegde, Shwetha K P,Samyak Katke, Chandresh Kumar Rastogi, Raja Vidya, and Manjunatha C*, Electrostatic Separation and AI-Driven Optimization for Lithium-Ion Battery Recycling:A Framework for a Sustainable Circular Economy, ACS Sustainable Resource Management , Manuscript ID: rm-2026-00014m , 2026,(Revision)
124	Quartz Tube-Assisted Sintering of LATP Solid Electrolyte for LFP Batteries: Suppressing Li ⁺ Loss and Enhancing Electrochemical Performance, Debabrata Mohanty, Jie-Yu Liao, Yun-Chien Yang, Somya Samantaray, Rinlee Butch M. Cervera, Girish Kumar S, Manjunatha C* , I-Ming Hung, Chemical Papers , 2026, (Under review)
123	Yashasvi Jaiswal, Chandresh Kumar Rastogi*, Ravi Maurya, C. Manjunatha , Gopal Ji, , Synthesis and application of bamboo leaves ethanol extract as an eco-friendly and corrosion-resistant coating for mild steel in saline solutions, Canadian Metallurgical Quarterly , Under review , 2026
122	KP Prabhakar Shwetha, Y Athreya, B M Rajesh, Chandresh K Rastogi, Sham Aan, MP, S. Girish Kumar, Manjunatha C* , Defect-Engineered Mg–Ce Co-Doped NiO Nanospheres for Enhanced Pseudocapacitive Energy Storage: Experimental and DFT Investigation, Journal of Materials Research , 2026 (Under review)
121	Shwetha KP, Rashmi Khan, Chandresh Kumar Rastogi, Sudha Kamath, Girish Kumar S, Radha N, Manjunatha C* , Experimental and DFT Insights into Mn-Doped NiS ₂ Nanostructures for High-Performance Supercapacitors, APL Energy , APE26-AR-00036, (Revision), 2026
120	R. Kavitha, C. Manjunatha , S. Girish Kumar, ZnO-based S-scheme heterojunction: Design principles, preparation methods and photocatalytic activity, Chinese Journal of Catalysis , Q-1 , IF= 17.7 83 (2026) 54–95, https://doi.org/10.1016/S1872-2067(26)64969-8
119	Debabrata Mohanty, Wei-Fan Kuan, K.P. Shwetha, I-Ming Hung, N. Radha, C. Manjunatha* , Facile and scalable synthesis of Li ₂ Mn(PO ₃) ₄ solid electrolyte for safe and stable Lithium batteries, Materials Letters , (Q-2, IF =2.9) 416, 2026, 140696, DOI: 10.1016/j.matlet.2026.140696.
118	Tailoring NiO-SrCO ₃ nanocomposites via fuel-controlled combustion synthesis for enhanced supercapacitor performance, Sultana, Zeba, Rastogi, Chandresh Kumar, Yadav, Neeraj Ji, Gopal, Girish Kumar, S. Shrivastava, Vishnu Prasad, Manjunatha, C* . New Journal of Chemistry (Q-2) , 2026, http://dx.doi.org/10.1039/D5NJ03043A
117	Ashwini Chavan V.M., Chandresh Kumar Rastogi, Shireesha Golla, Radha Nagaraj, Manjunatha Channegowda, Design and fabrication of efficient NiAl-LDH/CNT electrodes for high-performance supercapacitors with light-sensing functionality, Materials Science and Engineering: B, (Q-1) Volume 325, 2026, 119069, ISSN

	0921-5107, https://doi.org/10.1016/j.mseb.2025.119069 .
2025 Publications	
116	Neha Mathur, Monu Choudhary, Abhinav Kashyap Dwivedi, Jatin Nama, Shwetha KP, Manjunatha C , Sudhanshu Shama, Pankaj Gupta, Hemant Joshi, Partha Roy. "Versatility of Surfactant-Mediated NiTe ₂ Nanoparticles: Unlocking Potential for Hydrogen Evolution Reaction, Supercapacitor, and Sustainable Green Catalysis." Small (Q-1) 21, no. 42 (2025): e07377. https://doi.org/10.1002/sml.202507377
115	R. Kavitha, C. Manjunatha , Jiaguo Yu, S. Girish Kumar, Rational design and interfacial engineering of hierarchical S-scheme heterojunction and their photocatalytic applications, EnergyChem , 2025, 100159, ISSN 2589-7780, DOI: 10.1016/j.enchem.2025.100159. (Q-1, IF = 23.8, Cite Score : 40.1) , Indexed in WOS, SCI, and Scopus)
114	Naik, P. R., Rajashekara, V. A., Mudhulu, S., & C. Manjunatha* . Uranium removal from water by adsorption: a detailed characterisation and parametric study using nickel ferrite (NiFe ₂ O ₄) nanoadsorbent. Environmental Pollutants and Bioavailability , 2025, 37(1). https://doi.org/10.1080/26395940.2025.2515293 , (Q-2), Indexed in WOS, SCI, and Scopus)
113	Ashwini Chavan V M, Shireesha G, Chandresh Kumar Rastogi, Manjunatha C* , Nickel Chromium-LDH@CNT Composite for Supercapacitors: Synergistic Conductive Network for Superior Energy Storage, Journal of Power Sources, (Q-1) 654 (2025) 237807 https://doi.org/10.1016/j.jpowsour.2025.237807 , Indexed in WOS, SCI, and Scopus)
112	Jaiswal, Y., Rastogi, C. K., Maurya, R., Manjunatha, C. , Ji, Gopal. Synthesis of watermelon seed extract and silica nanocomposites coatings and their utilization for the corrosion prevention of mild steel in saline solutions. Canadian Metallurgical Quarterly , (Q-2) , 2025, 1–16. https://doi.org/10.1080/00084433.2025.2502706 Indexed in WOS, SCI, and Scopus)
111	Poojashri Ravindra Naik, Vinod Alurdoddi Rajashekara, Sudeep Mudhulu, Manjunatha Channegowda* , Facile synthesis, characterisation and application of zinc ferrite in removal of uranium from water by adsorption, Journal of Contaminant Hydrology, (Q-1) Volume 273, 2025, 104583, ISSN 0169-7722, https://doi.org/10.1016/j.jconhyd.2025.104583 . Indexed in WOS, SCI, and Scopus)
110	S. Beena, Chandresh Kumar Rastogi, Yash Athreya, K.P. Shwetha, Nelsa Abraham, Suresh Babu Viswanathan, M.K. Sudha Kamath, C. Manjunatha* , Unveiling the supercapattery behaviour of novel kotoite-type Co ₃ (BO ₃) ₂ nanoparticles, Journal of Energy Storage, (Q-1) Volume 110, 2025 , 115299, ISSN 2352-152X, https://doi.org/10.1016/j.est.2025.115299 .) Indexed in WOS, SCI, and Scopus)
109	Ashwini Chavan V.M., G. Shireesha, Chandresh Kumar Rastogi, Manjunatha C* , Enhanced Electrochemical Performance of NiFe-LDH@CNT Nanocomposite for High-Energy Supercapacitors, ACS Applied Electronic Materials, (Q-1) ,

	https://doi.org/10.1021/acsaelm.4c01381 , 2025 , 7, 1, 48–6, Indexed in WOS, SCI, and Scopus)
2024 Publications	
108	K.P. Shwetha, M.K. Sudha Kamath, Chandresh Kumar Rastogi, Yash Athreya, Sooryadas Sudhakaran, C. Manjunatha *, Tailored carbon nanotube-CoS ₂ nanocomposites for the development of asymmetric supercapacitor coin cells demonstrating enhanced electrochemical properties, Journal of Energy Storage, (Q-1) Volume 97, Part A, 2024, 112699, ISSN 2352-152X, https://doi.org/10.1016/j.est.2024.112699 . Indexed in WOS, SCI, and Scopus)
107	Manjunatha C* , Chandresh Kumar R, Manmadha Rao B, Girish Kumar S, Varun S, Kartik R, Gyanprakash Maurya, Karthik B, Swathi C, Mohtada Sadrzadeh, Ajit Khosla "Advances in Hierarchical Inorganic Nanostructures for Efficient Solar Energy Harvesting Systems", Wiley, ChemSusChem , e202301755. (2024) https://doi.org/10.1002/cssc.202301755 , (Q-1, IF= 8.4, Cite Score = 15.5, Indexed in WOS, SCI, and Scopus) (COVER FEATURE ARTICLE)
106	Shwetha KP, Yash Athreya, Sudha Kamath MK*, Girish Kumar S, Manjunatha C* , Ajit Khosla, Development of NiS@f-MWCNT Nanocomposite Based High Performance Supercapacitor Coin Cell Prototype Device, (2023) Journal of Energy Storage, 75, 1, 2024, 109404 , https://doi.org/10.1016/j.est.2023.109404 (Q-1, IF= 9.4, Cite Score = 10.3, Indexed in WOS, SCI, and Scopus)
105	Hiremath, S., Mudhulu, S., Vidya, C., Chandraprabha, M. N., Moses, V., & Manjunatha, C.* . Green synthesized nano-TiO ₂ from Tamarindusindica leaf extract for photo-degradation of combination of dyes by suspension and immobilization on inert supports. Green Chemistry Letters and Reviews , (Q-1, IF = 5.8) 2024, 17(1). https://doi.org/10.1080/17518253.2024.2420711
104	Manjunatha C , Arpit Verma, Igra Arabia, Ujwal S Meda, Ishpal Rawal, Sarevesh Rustagi, Bal Chandra Yadav, Patrick Dunlop, Nikhil Bhalla and Vishal Chaudhary, "High Selectivity and Sensitivity through Nanoparticle Sensors for Cleanroom CO ₂ Detection" IOP Nanotechnology 35 (2024) 315501 (10pp), https://doi.org/10.1088/1361-6528/ad3fbf . (Q-2, IF= 3.5, Cite Score = 6.7, Indexed in WOS, SCI, and Scopus)
103	Raghavendra Garlapally, Niharika MP, B. Manmadha Rao*, C Manjunatha C* , B Venkateswarlu, NeellaNagarjuna, "Advances in Different Crystallization Techniques for Anodically Grown Amorphous TiO ₂ Nanotubes and its Applications", IOP Physica Scripta , 2024, 99, 6, 062002, https://doi.org/10.1088/1402-4896/ad3583 , (Q-2, IF= 2.9, Cite Score = 4.4, Indexed in WOS, SCI, and Scopus)
102	K. P. Shwetha, P. N. Amogh, S. Shrinath Vaidya, N. Hariprasad, S. Krishna and C. Manjunatha* , "Nanorobotics for Advancing Biomedicine: Progresses in Materials, Design, Fabrication, Opportunities and Applications," in IEEE Access , (Q-1) vol. 12, pp. 91664-91677, 2024, doi: 10.1109/ACCESS.2024.3416818

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40	Anil Subash S, Shubha MB, Yash N Athreya, Ajit Khosla, and Manjunatha C* , "Advances in Printable, Flexible and Transparent Graphene Photodetectors for Optoelectronics Applications", ECS Transactions , 107 (1) 18681-18695 (2022), [10.1149/10701.18681ecst ©The Electrochemical Society] (h-index =56, cite score = 1.1, Indexed in Scopus)
39	Saipriya L , Akepati Deekshitha , Shreya , Shubhika Verma , Swathi C , and Manjunatha C* , "Advances in Graphene Based MEMS and Nems Devices: Materials, Fabrication and Applications", ECS Transactions , 107 (1) 10997-11005 (2022), [10.1149/10701.10997ecst ©The Electrochemical Society] (h-index =56, cite score =

	1.1, Indexed in Scopus)
38	Srujana K, Ananya M, Keerthana PS, Rachana M, Ashmeera SH, and Manjunatha C* , "Advances in Graphene Based Electronic Skin and Their Healthcare Applications", ECS Transactions , 107 (1) 17659-17670 (2022), [10.1149/10701.17659ecst ©The Electrochemical Society] (, h-index =56, cite score = 1.1, Indexed in Scopus)
37	Aniket Chakraborty, Sudeep M, Vishal Chaudhary*, Manjunatha C* , Sujan Chakraborty, and Krishna M, "Advances and Prospects in Architecting SnO ₂ Based Nanosystems for Gas Sensing Applications", ECS Transactions , 107 (1) 14289-14303 (2022), 10.1149/10701.14289ecst ©The Electrochemical Society], (h-index =56, cite score = 1.1, Indexed in Scopus)
2021 Publications	
36	Rajani Katiyar, K R Usha Rani, TS Sindhu, HD Sneha Jain, Vidhyashree V, S Ashoka and Manjunatha C , "Design and development of electrochemical potentiostat circuit for the sensing of toxic cadmium and lead ions in soil" Engineering Research Express 3 (2021) 045026. DOI:10.1088/2631-8695/ac3637, (Q-2, Cite score=1.1, Indexed in WOS, SCI, and Scopus)
35	Sudeep M, Manjunatha C , Sham Aan M. P, Ashoka S, Suresh R, Ujwal S. M. Development of Nano Copper Sulfide (CuS) Structures for Electrochemical Detection of Vitamin C. Journal of nanostructures , 2021; 11(3):628-637. DOI: 10.22052/JNS.2021.03.020, (Q-3, Cite score=2.2, Indexed in WOS, SCI, and Scopus)
34	Selvaraj Manickam, Assiri Mohammed A, Rokhum Samuel Lalthazuala, Manjunatha C , Appaturi Jimmy Nelson, Murugesan Sepperumal, Bhaumik Asim, Subrahmanyam CH, "Solvent-free benzylic oxidation of aromatics over Cu(ii)-containing propylsalicylaldehyde anchored on the surface of mesoporous silica catalysts" Dalton Transactions , (2021), DOI: 10.1039/D1DT01760H. (Q-2, IF=4.57, Cite Score = 7.3, Indexed in WOS, SCI, and Scopus)
33	Ansari Sajid Ali, Manjunatha C* , Parveen Nazish, Shivaraj B W, Hari Krishna R. "Mechanistic insights into defect chemistry and tailored photoluminescence and photocatalytic properties of aliovalent cation substituted Zn _{0.94} M _{0.06-x} Li _x O (M: Fe ³⁺ , Al ³⁺ , Cr ³⁺) nanoparticles", Dalton Transactions , (2021), DOI: 10.1039/D1DT01706C. (Q-2, IF=4.57, Cite Score = 7.3, Indexed in WOS, SCI, and Scopus)
32	J Shivakumara*, Manjunatha C* , S Ashoka, R Hari Krishna, Manickam Selvaraj, Mika Sillanpää, "Enhancement of Photoluminescence of CdSiO ₃ :Eu ³⁺ phosphor using Na ⁺ and K ⁺ as Charge compensators", Chemical Physics , (2021), 551, 111319, DOI: 10.1016/j.chemphys.2021.111319. (Q-2, IF=2.348, Cite Score = 4.1, Indexed in WOS, SCI, and Scopus)
31	N. Srinivasa, P. Rangaswamy, R.T. Yogeeshwari, Manjunatha C , S. Ashoka, "Mesoporous LiTiPO ₄ F nanoparticles: A new stable and high performance bifunctional electrocatalyst for electrochemical water splitting", Surfaces and Interfaces , (2021), 25, 101188, DOI: 10.1016/j.surfin.2021.101188. (Q-1, IF=6.1, Cite

	Score = 5, Indexed in WOS, SCI, and Scopus)
30	Manjunatha C* , S Lakshmikanth, L Shreenivasa, N Srinivasa, S Ashoka*, BW Shivraj, Manickam Selvaraj, Ujwal Shreenag Meda, G Satheesh Babu, Bruno G. Pollet*, "Development of non-stoichiometric hybrid $\text{Co}_3\text{S}_4/\text{Co}_{0.85}\text{Se}$ nanocomposites for an evaluation of synergistic effect on OER performance". Surfaces and Interfaces (2021) 25, 101161, DOI: 10.1016/j.surfin.2021.101161, (Q-1, IF=6.1, Cite Score = 5, Indexed in WOS, SCI, and Scopus)
29	Srinivasa, N; Hughes, Jack; Adarakatti, Prashanth; Manjunatha C. ; Rowley-Neale, Samuel; Siddaramanna, Ashoka; Banks, Craig, "Facile Synthesis of Ni/NiO Nanocomposites: The Effect of Ni Content in NiO upon the Oxygen Evolution Reaction within Alkaline Media", RSC Advances (2021), 11, 14654-14664, DOI: 10.1039/D0RA10597J, (Q-2, IF=3.369, Cite Score = 5.9, Indexed in WOS, SCI, and Scopus)
28	Manjunatha C* , Novel bismuth oxy hydride chromate ($\text{HBi}_3(\text{CrO}_4)\text{O}_3$) nano-sheets/rods synthesized by one step one pot wet chemical method, IOP SciNotes , (2021), DOI: 10.1088/2633-1357/abf634, (Indexed in CNKI, Inspec, NASA Astrophysics Data System, Scite, Yewno)
27	V. Mamtha, H. N. Narasimha Murthy, V. Pujith Raj, Prashantha Tejas, C. S. Puneet, Achyutha Venugopal, Sham Aan Mankunipoyil, Manjunatha C. , "Electrospun PU/MgO/Ag Nanofibers for Antibacterial Activity and Flame Retardancy" Clothing and Textiles Research Journal . (2021). DOI:10.1177/0887302X211009479, (Q-1, IF=1.9, Cite Score = 2.9, Indexed in WOS, SCI, and Scopus)
26	Manjunatha C* , "Functionalized Photocatalytic Nanocoatings for Inactivating COVID-19 Virus Residing on Surfaces of Public and Healthcare Facilities", Coronaviruses (2021); 2(12): e070921191640. DOI:10.2174/2666796702666210222110403, (Q-4, Cite Score = 1.3, Indexed in Scopus, Dimensions, J-Gate and Scilit)
25	Manjunatha C* , "Design and Development of Smart and Safe Helmet for Inhibiting COVID-19 Virus Infection: A Simple Idea for Solving the Big Crisis", Coronaviruses (2021); 2(8): e250621189087. DOI:10.2174/2666796701999201211213112, (Q-4, Cite Score = 1.3, Indexed in Scopus, Dimensions, J-Gate and Scilit)
24	Sinchana Raj and Manjunatha C* , "Biomimicry: Recent Updates on Nanotechnology Innovations Inspired by Nature Creations", Current Nanoscience , (2021) 17(5). DOI:10.2174/1573413716999201127111149, (Q-3, IF=1.42, Cite Score = 3.1, Indexed in WOS, SCI, and Scopus)
23	Manjunatha C* , P Preran Rao, Parth Bhardwaj, H Raju and D Ranganath, "New insight into the synthesis, morphological architectures and biomedical applications of elemental selenium nanostructures", Biomedical Materials , (2021) 16, 022010, DOI: 10.1088/1748-605X/abc026, (Q-2, IF=4.1, Cite Score = 5.5, Indexed in WOS, SCI, and Scopus)

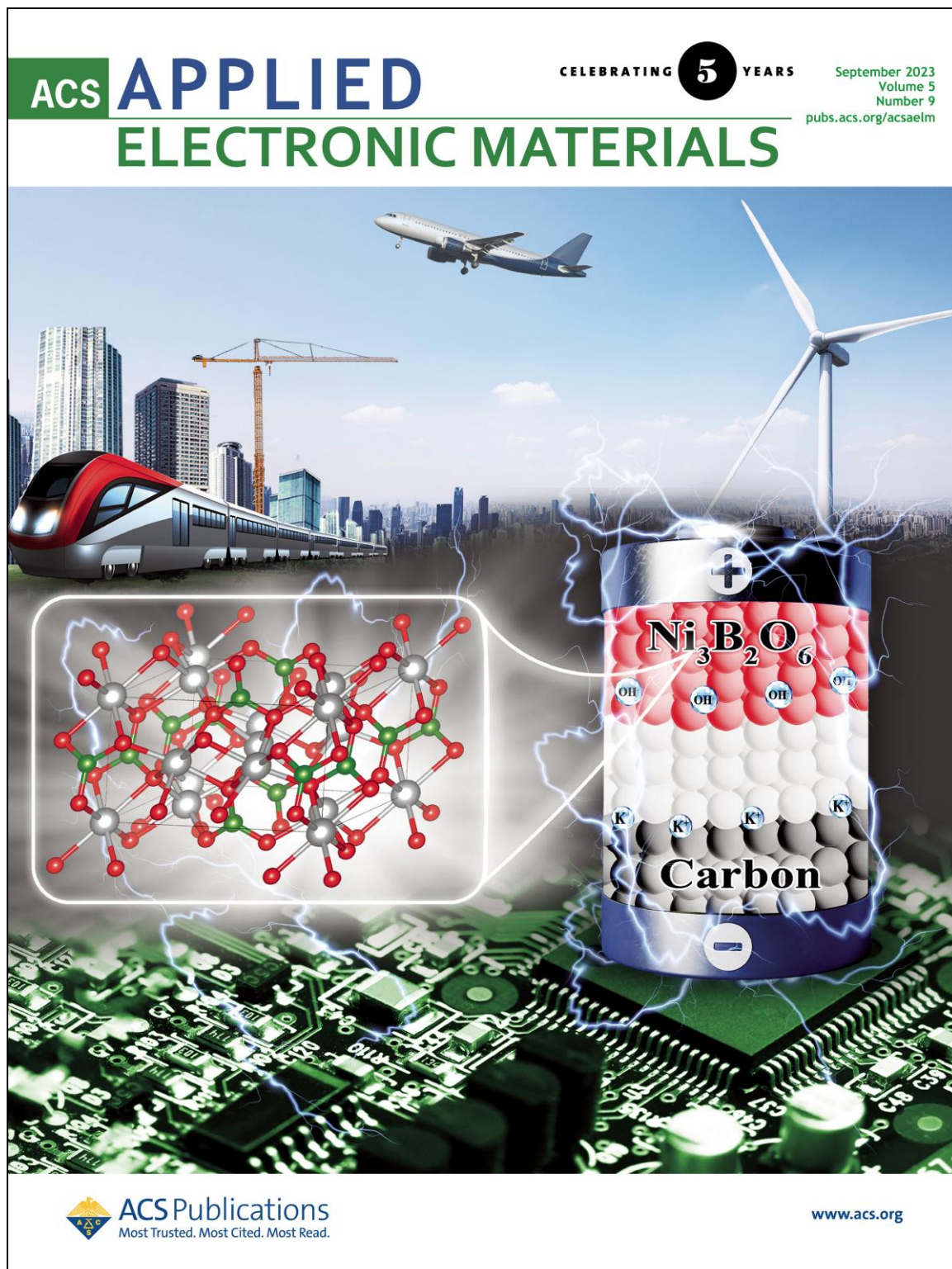
2020 Publications	
22	KR Usha Rani, Rajani Katiyar, Manjunatha C , Nivedita P Birajadar, Likhitha, Punith K. "Heavy metal-Ion Detection in soil using Anodic Stripping voltammetry", IEEE Xplore , International Conference for Emerging Technology (INCET) (2020), pp 473-476. DOI:10.1109/INCET49848.2020.9154169. (Indexed in Scopus)
21	Manjunatha C* , "Recent advances in environmentally benign hierarchical inorganic nano-adsorbents for the removal of poisonous metal ions in water: a review with mechanistic insight into toxicity and adsorption", Nanoscale Advances , (2020) 2, 5529-5554, DOI: 10.1039/d0na00650e, (Q-1, IF=5.59, Cite Score = 5.7, Indexed in WOS, SCI, and Scopus)
20	P.S. Subanaa, Manjunatha C* , B. Manmadha Rao, B. Venkateswarlu, G. Nagaraju, R. Suresh, "Surface functionalized magnetic α -Fe ₂ O ₃ nanoparticles: Synthesis, characterization and Hg ²⁺ ion removal in water", Surfaces and Interfaces , (2020) 21, 100680, DOI: 10.1016/j.surfin.2020.100680, (Q-1, IF=6.1, Cite Score = 5, Indexed in WOS, SCI, and Scopus)
19	Vijayakumar, B.W. Shivaraj, C. Manjunatha , B. Abhishek, G. Nagaraju, P.K. Panda, Hydrothermal synthesis of ZnO nanotubes for CO gas sensing, Sensors International , (2020), 1, 100018, DOI: 10.1016/j.sintl.2020.100018. (Q-1, Indexed in Scopus, WOS, SCI,
18	Manjunatha C* , Chirag V, Shivaraj BW, Srinivasa N, Ashoka S, "One pot green synthesis of novel rGO@ZnO nanocomposite and fabrication of electrochemical sensor for ascorbic acid using screen-printed electrode", Journal of Nanostructures , (2020) 10(3):9, DOI: 10.22052/JNS.2020.03.009, (Q-3, Cite Score=2.2, Indexed in WOS, SCI, and Scopus)
17	Manjunatha C* , N. Srinivasa, S. Samriddhi, C. Vidya, S. Ashoka, Studies on anion-induced structural transformations of iron(III) (Hydr)oxide micro-nanostructures and their oxygen evolution reaction performance, Solid State Sciences , (2020) 106, 106314, DOI:10.1016/j.solidstatesciences.2020.106314, (Q-2, IF=3.059, Cite Score =5.2, Indexed in WOS, SCI, and Scopus)
16	C Vidya, Manjunatha C* , Sudeep M, Ashoka S, Lourdu Antony Raj M A, "Photo-assisted mineralization of Titan Yellow dye using ZnO nanorods synthesized via environmental benign route" Discover Applied Sciences (formerly SN Applied Sciences) , (2020) 2, 743, DOI:10.1007/s42452-020-2537-2, (Q-2, Cite Score =2.7, Indexed in WOS, SCI, and Scopus)
15	Manjunatha C* , N. Srinivasa, S.K. Chaitra, M. Sudeep, R. Chandra Kumar, S. Ashoka, "Controlled synthesis of nickel sulfide polymorphs: studies on the effect of morphology and crystal structure on OER performance", Materials Today Energy , (2020) 16, 100414, DOI: 10.1016/j.mtener.2020.100414. (Q-1, IF=9.25, Cite Score = 10.4, Indexed in WOS, SCI, and Scopus)
14	Manjunatha C * , B. Abhishek, B. W. Shivaraj, S. Ashoka, M. Shashank, G. Nagaraju,

	Engineering the $M_xZn_{1-x}O$ ($M = Al^{3+}, Fe^{3+}, Cr^{3+}$) nanoparticles for visible light-assisted catalytic mineralization of methylene blue dye using Taguchi design, Chemical Papers , (2020), 74, 2719–2731 DOI: 10.1007/s11696-020-01113-5. (Q-2, IF=2.1, Cite Score = 3.1, Indexed in WOS, SCI, and Scopus)
13	Manjunatha C * , Rahul S Patil, M. Sudeep, N. Srinivasa, R. Chandra Kumar, M.P. Sham Aan, S. Ashoka, "Rational design and synthesis of hetero-nanostructured electrospun PU@PANI@FeS ₂ : A surface tailored hybrid catalyst for H ₂ production via electrochemical splitting of water", Surfaces and Interfaces , (2020), 18, 10445, DOI:10.1016/j.surfin.2020.100445, (Q-1, IF=6.1, Cite Score = 5, Indexed in WOS, SCI, and Scopus)
2019 Publications	
12	M. Jena, Manjunatha C * , B W Shivraj, G Nagaraju, S. Ashoka, M. P. Sham Aan, "Optimization of parameters for maximizing photocatalytic behaviour of $Zn_{1-x}Fe_xO$ nanoparticles for methyl orange degradation using Taguchi and Grey relational analysis Approach", Materials Today Chemistry , (2019), 12, 187-199. DOI: 10.1016/j.mtchem.2019.01.004, (Q-1, IF=7.6, Cite Score = 8.8, Indexed in WOS, SCI, and Scopus)
11	G K Nagendra, B W Shivaraj, C. Manjunatha, S A Ayesha Siddiqua and V Suchithra "Study of structural features and antibacterial property of ZnO/CuO nanocomposites derived from solution combustion Synthesis" IOP Conference Series: Materials Science and Engineering , 577, (2019), 012111, DOI:10.1088/1757-899X/577/1/012111. (Cite Score = 1.1 , Indexed in Scopus).
2018 Publications	
10	G. Shireesha, Manjunatha C , Anjana Jain, M.C.Radhakrishna, "Influence of $Cd_{0.99}Eu_{0.01}SiO_3$ nanoparticles concentration on $Cd_{0.99}Eu_{0.01}SiO_3$ /PVDF nanocomposite films", MATERIALS TODAY: proceedings . (2018) 5, 10, Part 1, Pages 21162-21174, DOI: 10.1016/j.matpr.2018.06.515, (Cite score =2.3, Indexed in Scopus)
9	J. Shivakumara Chikkahanumantharayappa S.Ashoka R. Hari Krishna, G.Vijayakumar, Manjunatha C , B.M.Nagabhushana, G.Nagaraju, "Elimination of quenching defects by facile anion doping in $CdSiO_3$ synthesized by green fuel assisted combustion method" Optik - International Journal for Light and Electron Optics , (2018), 154, 670-675, DOI:10.1016/j.ijleo.2017.10.066, (Q-2, IF=2.84, Cite Score =4.8, Indexed in WOS, SCI, and Scopus)
2017 Publications	
8	C Vidya, Manjunatha C * , Chandrababha. M.N, Megha Rajshekar, Antony Raj.M.A.L, "Hazard free green synthesis of ZnO nano-photo-catalyst using Artocarpus Heterophyllus leaf extract for the degradation of Congo red dye in water treatment applications" Journal of Environmental Chemical Engineering , (2017), 5, (4), 3172-3180. DOI: 10.1016/j.jece.2017.05.058, (Q-1, IF=7.97, Cite Score = 7.7, Indexed

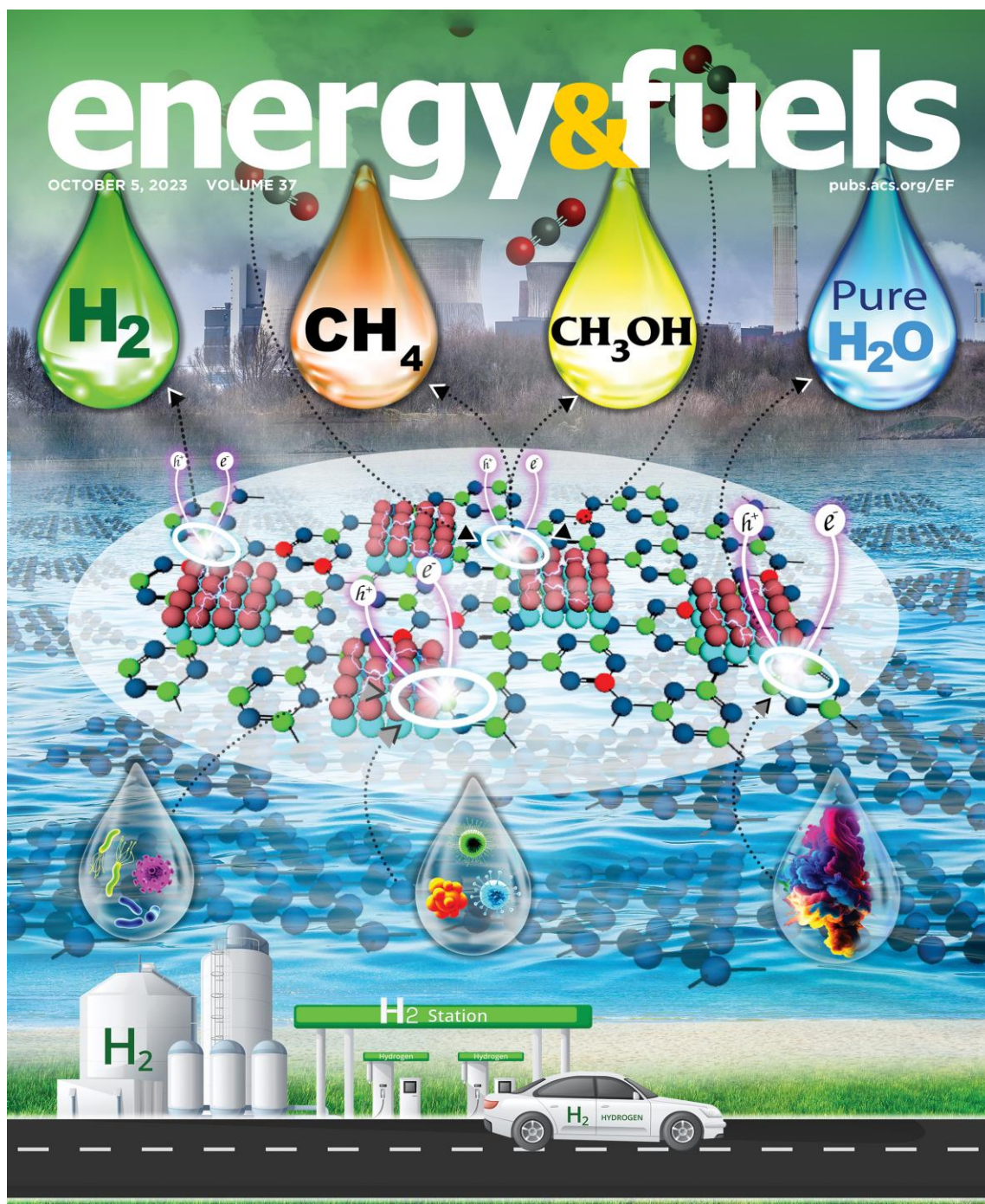
	in WOS, SCI, and Scopus)
2012-2013 Publications (PhD Publications)	
7	Manjunatha C , D.V. Sunitha, H. Nagabhushana, Fouran Singh, S.C. Sharma, R.P.S. Chakradhar and B. M. Nagabhushana, "Thermoluminescence properties of 100 MeV Si ⁷⁺ swift heavy ions and UV irradiated CdSiO ₃ :Ce ³⁺ nanophosphor", Journal of Luminescence , (2013) 134 , 358-368. DOI: 10.1016/j.jlumin.2012.08.020, (Q-2, IF=4.17, Cite Score = 6.9, Indexed in WOS, SCI, and Scopus)
6	Manjunatha C , B.M. Nagabhushana, D.V. Sunitha, H. Nagabhushana, S.C. Sharma and R.P.S. Chakradhar, "Influence of halide flux on the crystallinity, microstructure and thermoluminescence properties of CdSiO ₃ :Co ²⁺ nanophosphor" Materials Research Bulletin , (2013) 48 , 158-166. DOI: 10.1016/j.materresbull.2012.09.068, (Q-1, IF=5.6, Cite Score =9.8, Indexed in WOS, SCI, and Scopus)
5	Manjunatha C , B.M. Nagabhushana, D.V. Sunitha, H. Nagabhushana, S.C. Sharma, G. B. Venkatesh and R.P.S. Chakradhar, "Effect of NaF flux on microstructure and thermoluminescence properties of Sm ³⁺ doped CdSiO ₃ phosphor", Journal of Luminescence , (2013), 134 , 432-440. DOI: 10.1016/j.jlumin.2012.08.006, (Q-2, IF=4.17, Cite Score = 6.9, Indexed in WOS, SCI, and Scopus)
4	Manjunatha C , D.V. Sunitha, H. Nagabhushana, B.M. Nagabhushana, S.C. Sharma, and R.P.S. Chakradhar, "Combustion synthesis, structural characterization, thermo and photoluminescence studies of CdSiO ₃ :Dy ³⁺ nanophosphor", Spectrochimica Acta Part A , (2012) 93 , 140–148. DOI: 10.1016/j.saa.2012.02.094, (Q-2, IF=4.831, Cite Score = 7.1, Indexed in WOS, SCI, and Scopus)
3	Manjunatha C , D.V. Sunitha, H. Nagabhushana, S.C. Sharma, S. Ashoka, J.L. Rao, B.M. Nagabhushana and R.P.S. Chakradhar, "Structural characterization, EPR and thermoluminescence properties of Cd _{1-x} Ni _x SiO ₃ nanocrystalline phosphors", Materials Research Bulletin , (2012) 47 , 2306–2314. DOI: 10.1016/j.materresbull.2012.05.039, (Q-1, IF=5.6, Cite Score = 9.8, Indexed in WOS, SCI, and Scopus)
2	D.V. Sunitha, Manjunatha C , C.J. Shilpa, H. Nagabhushana, S.C. Sharma, B.M.Nagabhushana, N. Dhananjaya, C. Shivakumara and R.P.S. Chakradhar, "CdSiO ₃ :Pr ³⁺ nanophosphor: Synthesis, characterization and thermoluminescence studies" Spectrochimica Acta Part A , (2012) 99 , 279–287. DOI: 10.1016/j.saa.2012.08.057, (Q-2, IF=4.831, Cite Score = 7.1, Indexed in WOS, SCI, and Scopus)
1	Manjunatha C* , B.M. Nagabhushana, H. Nagabhushana and R. P. S. Chakradhar, "Transformation of hydrothermally derived nanowire cluster intermediates into CdSiO ₃ nanobelts", Journal of Materials Chemistry-A (formerly Journal of Materials Chemistry) , 22 , (2012) 22392-22397. DOI: 10.1039/C2JM34356H, (Q-1, IF=14.51, Cite Score = 21, Indexed in WOS, SCI, and Scopus)

FRONT COVER PAGE ARTICLES

1. <https://pubs.acs.org/toc/aaembp/5/9>



2. <https://pubs.acs.org/toc/enfuem/37/19>



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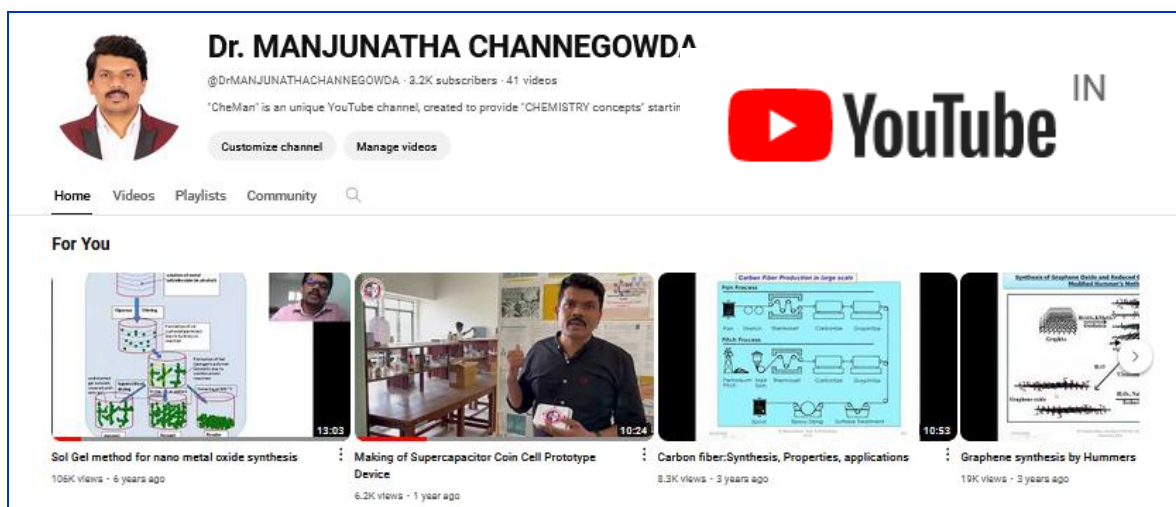
C. Manjunatha, C. K. Rastogi, A. Khosla and co-workers

Advances in Hierarchical Inorganic Nanostructures for Efficient Solar Energy Harvesting Systems



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Book Chapters Published:

- 1. Title of the Book:** GREEN SUSTAINABLE PROCESS FOR CHEMICAL AND ENVIRONMENTAL ENGINEERING AND SCIENCE: Green Inorganic Synthesis
Title of book chapter: Microwave Assisted Green Synthesis of Inorganic Nanomaterials
Authors: **C. Manjunatha ***, Ashoka S, Hari Krishna R
Editors: Inamuddin Rajender Boddula, Mohd Imran Ahamed, Abdullah M. Asiri
Publishers: Elsevier, Paperback ISBN: 978-0-12-821887-7, eBook ISBN: 978-0-12-821901-0, DOI: 10.1016/B978-0-12-821887-7.00011-2, published date 1st December 2020. Page 1-40.
- 2. Title of the Book:** GREEN SUSTAINABLE PROCESS FOR CHEMICAL AND ENVIRONMENTAL ENGINEERING AND SCIENCE: Green Inorganic Synthesis
Title of book chapter: Green synthesis of inorganic nanoparticle using microemulsion methods
Authors: **C. Manjunatha ***, Ashoka S, Hari Krishna R
Editors: Inamuddin Rajender Boddula, Mohd Imran Ahamed, Abdullah M. Asiri
Publishers: Elsevier, Paperback ISBN: 978-0-12-821887-7, eBook ISBN: 978-0-12-821901-0, DOI: 10.1016/B978-0-12-821887-7.00002-1, published date 1st December 2020. Page 1-40.
- 3. Title of the Book:** Essential Oils: Extraction Methods and Applications
Title of book chapter: Supercritical CO₂ Extraction as a Clean Technology Tool for Isolation of Essential Oils
Authors: T. P. Krishna Murthy, R. Hari Krishna, M. N. Chandra Prabha, Priyadarshini Dey, Blessy Baby Mathew, **C. Manjunatha**
Editor: Inamuddin

Publishers: Wiley- Scrivener, 2023, Print ISBN:9781119829355, Online ISBN:9781119829614 DOI:10.1002/9781119829614, <https://doi.org/10.1002/9781119829614.ch34>

Membership of Technical Bodies

1. Member of American Chemical Society (ACS, USA, 31610891)
2. Member of Royal Society of Chemistry (RSC, UK) (730705)
3. Member of Electrochemical Society (ECS)-USA (504170)
4. Indian Society for Technical Education (ISTE); ID no.: LM 105901
5. Luminescence Society of India: Karnataka Chapter (LSI-KC): ID No: LM 010.

Funded Projects/Fellowships:

Sl. No	Project Title	Funding Agency	Total Amount Sanctioned	Role	Year	Status
1	Establishment of Centre of Excellence in "Nanomaterials and Devices" (CND)	RSST, Bengaluru	Rs. 30 Lakhs	PI	2022	On-Going
2	Establishment of Bio-fuel Research Information and Demonstration Centre (BRIDC-RVCE)" at RV College of Engineering	Karnataka State Bio-fuel Development Board (KSBDDB), Government of Karnataka	Rs. 75 Lakhs	Co-PI	2012	On-Going
3	Consultancy - Research Idea, experimentation, execution, paper/patent publication	PhD scholar from KTU, Kerala	Rs 1.5 Lakhs	PI	2023	On going
4	Seed Money Grant	Rashtreeya Sikshana Samithi Trust (RSST) Bengaluru, India under RVCE sustainability fund. No:RVE/A/c/116/2021-22/ Dated 08 July 2021	Rs. 1.35 Lakhs	PI	2021	On-Going
5	International Travel Support (ITS)	Science & Engineering Research Board (SERB),	Rs 0.42827 Lakhs	PI	2019	Completed

		Department of Science and Technology, Government of India Sanction Order : ITS/2019/000577				
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Supervising Ph.D Students:

Visvesvaraya Technological University, Belagavi-590018, Karnataka, INDIA.

Sl. No	Name of the student (USN)	University -Discipline	Thesis title	Registration year	Status
1	Shubha M.B (1RV18PCT02)	VTU- Applied Sciences (Chemistry)	Rational design and synthesis of hierarchical metal chalcogenide nanomaterials for optoelectronic devices	2017	Course work completed, One Q1 Paper and two Q4, published as first author
2	Anil Subash. S (1RV19PCT01)	VTU- Applied Sciences (Chemistry)	Development of functionalized graphene anchored with 0-3 dimensional inorganic nanomaterials for sensor applications	2018	Course work completed, One Q1 Paper and two Q4, published as first author

UG/PG (B.E/M.Tech) Students projects guided/guiding- 62 projects, 150 students.

2022-25 UG-PG Research Projects	
62	Synthesis and characterisation of transition metal oxide nanomaterials and electrospun fibers for supercapacitor applications Manjunatha C*, Navaneeth M (Consultancy-External student) (MZAVMCH019) M.Sc Chemistry, (Markaz Arts & Science College, Kerala)
61	Development of Divalent Metal Borates for High Performance Supercapacitors Manjunatha C*, Yashesh Vijay Rajyaguru (1RV19CH029), Shravan S Ranga (1RV19CH034), Vishwesh S Desai (1RV19CH037), Sharva Jois (1RV19CH032), 8th Sem BE-Chemical, UG-Major project.
60	Development of Bimetallic selenides-based Nanostructures for supercapacitor applications Manjunatha C*, Shamanth S Nadig (1RV21MPD19) M.Tech (PDM) PG-Major project ME
59	Development of Transition Metal Oxide Nanostructure For Biosensor Electrode Manjunatha C*, Srujan B J (1RV21MPD03), M.Tech (PDM) PG-Major project ME

58	Development Of Nickel Based Chalcogenides For Hydrogen Generation Manjunatha C*, Vishal S Bhagini (1RV21MPD27) M.Tech (PDM) PG-Major project ME
57	Synthesis and charectriation of Nckel bismuth oxide nanoparticles for gas sensor applications Manjunatha C*, Naveen Kumar M.Tech (PDM) PG-Major project ME
56	Development of Cerium oxides nanoparticles for EC biosensors Manjunatha C*, Basava Prabhu S H, (Consultancy-External student), (1BY21EC027) BE III SEM (BMSITM, Bengaluru)
55	Development of Doped NiO nanoparticles for EC biosensors Manjunatha C*, Anchal Jain, (Consultancy-External student) (JS211772) M.Sc (NST) JSSAHER Mysuru
54	Development of Pure NiO nanoparticles for EC biosensors Manjunatha C*, Deepthi GV, (Consultancy-External student) (JS211770), M.Sc (NST), JSSAHER Mysuru
53	Major projectDevelopment of Bimetallic oxides for electrochemical sensor applications Manjunatha C*, Vishnu Vardhan (1RV17CH042), Chalichama (1RV19CH007), Arko Bose (1RV19CH025), Pasupuleti Venkata Sai Dinesh (1RV20CH401) 8th Sem BE-Chemical UG- Major project.
52	Study the electrochemical sensing property of nanomaterials decorated screen printed and glass carbon electrode Manjunatha C*, Rakshita GY (1RV21EC134) BE (ECE) III Sem, Internship
51	Study the sensor property of commercial Bi ₂ O ₃ for electrochemical uric acid sensor applications Manjunatha C*, Manish Danda (1RV21BT027), BE (BT) III Sem, Internship
50	Development of bismuth oxide for electrochemical ascorbic acid sensor applications Manjunatha C*, Iqra Noorin (1RV21BT020) BE (BT) III Sem, Internship
49	Study the sensor property of commercial V ₂ O ₅ for electrochemical uric acid sensor applications Manjunatha C*, Sumedh S C (1RV21BT053) BE (BT) III Sem, Internship
48	Development of Nickel cobalt oxide for electrochemical glucose sensor applications Manjunatha C*, Anirudh R Urs (1RV21BT007) BE (BT), III Sem, Internship
47	Development of Nickel vanadate for electrochemical glucose sensor applications Manjunatha C*, Sukruth Kashyap S (1RV21BT052) BE (BT), III Sem, Internship
46	Synthesis of NiS for supercapcitors applications Manjunatha C*, Dinesh PVS 1RV19CH025 BE (CHE), III Sem, Internship
45	Fabrication of Cobalt sulphide based coin cell type supercapacitor Manjunatha C*, Meeth M Parekh(RVCE19BME030) BE (ME) III Sem,Internship
2021-22 UG-PG Research Projects	
44	Synthesis of Nano-Metal Oxide for Enhancing Properties of Cement Concrete,

	Manjunatha C*, Channabasava (1RV18CV028), Chirag Jain V (1RV18CV027), Gundappa (1RV18CV037), Mahesh Basavaraj (1RV18CV054), 8th Sem BE-CIVIL UG-Major project
43	Development of low cost eco-friendly nano-silicate additives for the enhancement of concrete properties. Manjunatha C*, Umar Bashir (1RV18CV119), Sagar M (1RV19CV408), Shashank Rayaraddi (1RV18CV099), Manjunath Bhavi (1RV18CV056), 8th Sem BE-CIVIL UG-Major project
42	Investigation of the effect of functionalized CNT as positive electrode material in lead-acid battery Manjunatha C*, Aneesh Bhat P (1RV18ME017), Arun Kumar B (1RV18ME024), Tulasiram TS (1RV18ME114), Vikas S (1RV18ME119), 8th Sem BE-ME UG-Major project
41	Synthesis and Characterization of Transition Metal Sulphide nano structure for Supercapacitor applications Manjunatha C*, Anuj R Kengunte (1RV18ME021), Avaneesh B Ballal (1RV18ME031), Girish B Umadi (1RV18ME047), Shubham Bhogan (1RV19ME409), 8th Sem BE-ME UG-Major project
40	Development of Hybrid Metal Chalcogenides for High Performance Supercapacitors Manjunatha C*, Yash Athreya (1RV18CH041), Shweta A Ram (1RV18CH027), Suraj L (1RV18CH030), Ananda Islakar (1RV18CH002), 8th Sem BE-Chemical UG-Major project
39	Development of NiS/NiO hybrid nanoelectrocatalysts formed from NiS for hydrogen production Manjunatha C*, Kiran N K (1RV20MPD13), M.Tech (PDM) PG-Major project ME
38	Development of WO ₃ /MWCNT Nanoparticles electrode for NO _x senso Manjunatha C*, Harshith Raj A (1RV20MPD10) M.Tech (PDM) PG-Major project ME
37	Optimization of CuX (X = Se, S, O) Nano structures for Vitamin C sensor Manjunatha C*, Naveen Kumar A (1RV20MPD18) M.Tech (PDM) PG-Major project ME
36	Development of Tungsten sulfide/Tungsten selenide nanocomposite electrode for water splitting Manjunatha C*, Baqar Ali Quraishi (1RV20MCM01) M.Tech (MCM), PG-Major project ME
35	Development of Nickel based Chalcogenides for Hydrogen generation Manjunatha C*, Sujay K M (1RV20MPD29) M.Tech (PDM), PG-Major project-ME
34	Development of RGO/Iron oxide composite for electro-chemical biosensor applications Manjunatha C*, Naveen R G (1RV20MMD09), M.Tech (MMD) PG-Major project ME
33	Development of CNT/Metal oxide Nanocomposite Supercapacitor Manjunatha C*, Vinaykumar (1RV20MMD15), M.Tech (MMD) PG-Major project

	ME
32	Development and Characterization Of CuO/ZrO ₂ Hybrid Nanocomposite Lubricant, Manjunatha C*, Sourabh Mohrir (1RV20MPD28), M.Tech (PDM) PG-Major project ME
31	Development of boron nitride coated fabric for flame retardant application Manjunatha C*, Raghav Srinath AK (1RV16ME083), Narendra RV (1RV17ME411), Kumar Ujwal (1RV16ME050), Laksmish S Hegde (1RV16ME052), 8th Sem BE-ME UG- Major project
30	Study of influence of CuO and palm oil milled nano cutting fluid on machining performance of SS-304 alloy. Manjunatha C*, Abhishek Holkar (1RV17ME003), Mahadev J (1RV17ME056), Mukund Achar (1RV17ME062), Suvijit Chakraborty (1RV17ME114), 8th Sem BE-ME-UG-Major project
29	Design of signal conditioning circuit for heavy metal ion detection in soil samples Manjunatha C*, Sindhu TS (1RV18EC430), Sneha Jain HD (1RV18EC431), Vidyashree V 1RV18EC433, 8th Sem BE-ECE, UG-Major project
28	Development of printed CNT/metal oxide nanocomposite thin film for sensor and device applications Manjunatha C*, Mahantesh Magi (1RVEC17068), Rahul MG (1RVEC17115), Sammed Endoli (1RVEC17135), Srujan R Rajanalli (1RVEC17165), 8th Sem BE-ECE UG-Major project
27	Development of Bi ₂ MO ₆ (M=Mo, W) based Z-scheme catalyst for photo/electrochemical applications Manjunatha C*, Tejas Patha Rao (1RV16CH033), Gokul Prakash (1RV17CH012), Gnana Soundarya V (1RV17CH011), 8th Sem BE Chemical UG-Major project
26	Synergistic effect of CuS@CuSe hybrid composite on Photo/Electro Catalytic application Manjunatha C*, Suryajeet Patil Nikam (1RV19MPD32) M.Tech (PDM) PG-Major project ME
25	Development of α -Fe ₂ O ₃ decorated electrode for electrochemical Glucose sensor application, Manjunatha C*, Md Saad Patel (1RV19MPD21) M.Tech (PDM), PG-Major project-ME
24	Development of nanoscale ITO ink for transparent conducting electrode Manjunatha C*, Preetam Kumar Hiroli (1RV19MPD26), M.Tech (PDM) PG-Major project-ME
23	Synthesis and characterisation of Cu, Zn, Co doped NiO nanoparticles for methane sensing application. Manjunatha C*, Vijay Kumar P (1RV19MPD35), M.Tech (PDM) PG-Major project-ME
22	Development of Selenium/Titanium dioxide nanocomposite for CO gas sensor Manjunatha C*, B Varun Kumar (1RV19MPD07) M.Tech (PDM), PG-Major project-ME
21	Development of RGO/CNT decorated metal oxide nanocomposite for methane gas

	sensing application, Manjunatha C*, Aniket Chakraborty (1RV19MPD05) M.Tech (PDM) PG-Major project ME
20	Surface Functionalised Hierarchical MoS ₂ Nanostructure for Water Splitting applications Manjunatha C*, Abdul Rahman Faisal (1RV19MPD02) M.Tech (PDM), PG-Major project-ME
2019-20 UG-PG Research Projects	
19	Development of iron oxide nano fluid for enhancement of heat transfer in a concentric heat exchanger Manjunatha C*, M Raghva Reddy (1RV13ME057), Deepak JB (1RV17ME401), Keishna Akami (1RV17ME407), Naresh (1RV17ME413), 8th Sem BE ME UG-Major project
18	Design of Electrochemical system for the detection of heavy metals in soil using anodic stripping voltammetry Manjunatha C*, Likitha (1RV16EC082), Nivedita P B (1RV16EC189), Punith K (1RV17EC422), 8th Sem BE UG-Major project-ECE
17	Development of Metal Sulfide/Metal Selenides Hybrid Nanostructure for Water Splitting Application Manjunatha C*, Lakshmikanth, (1RV18MPD11), M.Tech (PDM) PG-Major project-ME
16	Development and Characterization of Nanohybrid Composites of Vanadium Oxide/reduced Graphene oxide for Ethanol Gas sensing Applications. Manjunatha C*, Nanagouda Patil (1RV18MPD16) M.Tech (PDM), PG-Major project- ME,
15	Design and Development Of Nano Structured Metal Selenide Embedded Screen Printed Electrode For Biomolecule Sensors Manjunatha C*, Mohammed Asif (1RV18MPD13), M.Tech (PDM), PG-Major project-ME
14	Green synthesis of Selenium nanoparticles for biomedical applications, Manjunatha C*, P. Preran Rao (1RV16CH014), Parth Bhardwaj (1RV15CH043), 8th Sem BE UG-Major project
13	Green synthesis of transition metal doped MoS ₂ nanoparticles for electrochemical water splitting, Manjunatha C*, Faraaz Mohammed Ikram (1RV16CH008), Ashish John Thomas (1RV16CH005), 8th Sem BE UG-Major project-CE
2018-19 UG-PG Research Projects	
12	Tailored design and fabrication of surface engineered magnetic membrane for removal of toxic metal ions in water, Manjunatha C*, Samriddhi Saxena (1RV15CH024), Swizel Concy Pereira (1RV15CH037), Shivam Bansal (1RV15CH029), Ashwin Mathew (1RV15CH005), Pranjal Anand (1RV16CH015), 8th Sem BE UG-Major project-CE
11	Optimisation of process parameters of ZnO nanotubes for CO gas sensing, Manjunatha C*, Vijayakumar Lamani (1RV17MPD35) M.Tech (PDM), PG-Major project-ME
10	Synthesis of AZO/FZO/CZO nanoparticles for methylene blue dye degradation, Manjunatha C*, Abhishek. B (1RV17MPD01) M.Tech (PDM), PG-Major

	project-ME
9	Design and optimisation of parameters for the development of hierarchical metal dichalcogenide nano structures, Manjunatha C*, Chaitra SK (1RV17MPD11) M.Tech (PDM), PG-Major project-ME
8	Graphene/metal dichalcogenide composites: synthesis and electrochemical applications, Manjunatha C*, Shahrukh Khan Kittur (1RV17MPD32) M.Tech (PDM), PG-Major project ME
7	Tailored design and fabrication of electrospun ironsulfide/polymer nanofibers for solar energy-assisted electrochemical splitting of water , Manjunatha C*, Patil Rahul Shankar, (1RV17MPD22) M.Tech (PDM) PG-Major project ME,
6	Electrospun MoS ₂ /Polymer Nanofibers: Rational Design and Fabrication for Electrochemical Sensing Applications Manjunatha C*, Mohammed Mukarram Ahmed Ansari (1RV17MPD21), M.Tech (PDM), PG-Major project ME
2017-18 UG-PG Research Projects	
5	Synthesis and optimization of Zn _{1-x} Fe _x O nano-photocatalyst for methyl orange dye degradation", Manjunatha C*, Mrutunjaya Jena (1RV16MPD15), MTech (ME), VTU*-Major Project,
4	Synthesis, Characterisation and testing of ZnO/CuO nanocomposites for antibacterial application, Manjunatha C*, Nagendra G K (1RV16MPD16) M.Tech (ME) VTU-Major Project
3	Synthesis and characterization of thiol-functionalised core shell magnetic iron oxide nano-particles for Hg ²⁺ ion removal in water, Manjunatha C*, Sagar R Kokatnur, (1RV16HCE14) M.Tech (CE), VTU-Major Project
2	Synthesis and characterisation of Al doped ZnO for dye degradation, Manjunatha C*, Koushik H M (1RV17MPD18), Abhishek B (1RV17MPD01), M.Tech (ME) VTU-Minor Project
1	Development of Inorganic nanostructures for removal of organic pollutants in waste water, Manjunatha C*, Suhas D Bharadwaj (1RV12CH033), Shubham Gangwal (1RV12CH029), B.E (CE) VTU-Major Project
	* VTU-Visvesvaraya Technological University , ME : Mechanical Engineering, CE: Chemical Engineering

RESEARCH PAPERS PRESENTED IN INTERNATIONAL CONFERENCES

1. **C. Manjunatha**, H. Nagabhushana, R.P.S. Chakradhar and B.M. Nagabhushana, "Morphological Evolution of Cadmium Silicate Nanobelts". **International**

conference on Nano Science and Nano Technology [**ICONSAT-2012, page no. 287**] organized by ARCI, Hyderabad, India on 20-23, January-2012.

2. **C. Manjunatha**, H. Nagabhushana, R.P.S. Chakradhar and B.M. Nagabhushana "Characterization, luminescence studies of combustion derived Dy^{3+} doped CdSiO_3 nanophosphor", **International Conference** on Luminescence and its Applications (**ICLA-2012, page no. 285**) Organized by Rajiv Gandhi University of Knowledge Technologies, Hyderabad, India , Indian Institute of Chemical Technology, Hyderabad, India, Society for Information Displays (India Chapter), and Luminescence Society of India, on 7-10th February, 2012, Hyderabad, India.
3. **C. Manjunatha**, B.M. Nagabhushana, H. Nagabhushana, R.P.S. Chakradhar, "Enhancement of thermoluminescence property of $\text{CdSiO}_3:\text{Sm}^{3+}$ nanophosphor by flux aided combustion process for radiation dosimetry". **International conference** on "Laser Advances in Science Engineering and Research: (**ICLASER-2012, Page No. 44**)", June 8-10, 2012, M.S. Ramaiah Institute of Technology, Bangalore, India.
4. **C. Manjunatha**, B.M. Nagabhushana, presented the work on "Facile Synthesis of CdSiO_3 Nanobelts: A Versatile Luminescent Host" at **Winter School-2012 on Frontiers in Materials Science**, conducted by Jawaharlal Nehru Centre for Advanced Scientific Research (**JNCASR**), Bangalore 560064 India in association with **University of Cambridge**, Department of Materials Science and Metallurgy, Pembroke Street, **Cambridge CB2 3QZ**, and **Ras Al Khaimah Center for Advanced Materials**, Ras Al Khaimah, **United Arab Emirates** at JNCASR from 3rd to 8th December 2012.
5. **C. Manjunatha**, "Preparation and characterisation af PVDF- CdSiO_3 nanocomposite thin film" is presented in the International Conference on Advanced Materials, Manufacturing, Management and Thermal Sciences (AMMMT 2013) held at Siddaganga Institute of Technology, Tumkur during 3-4 may 2013.
6. G.Shireesha, **C. Manjunatha**, Anjana Jain, M C Radhakrishna Participated in First International conference on " Large Area and Flexible Micro Electronics (ILAFM-2014), Organised by Centre of Excellance (TEQIP) Macroelectronics during 18-20 Dec 2014, at R.V. College of Engineering, Bangalore-59. (Investigation on Characteristics of PVDF/ $\text{Cd}_{0.99}\text{Ni}_{0.01}\text{SiO}_3$ films for high-k capacitors, , International Conference on Large Area Flexible Macro Electronics-2014, R V College of Engineering, Bangalore, 18th-20th Dec.2014)
7. **C. Manjunatha**, Presented the research work entitled " Effect of NaCl flux on Photoluminescence of Solution combustion derived $\text{CdSiO}_3:\text{Ce}^{3+}$ nanoparticles., at Second Indo-Canadian Symposium on Nano-Science and Technology (ICSNST-216),

organized by The National Institute of Engineering, Mysuru, Karnataka, India in association with McMaster University, Hamilton, Canada during 18-19 February 2016.

8. **C. Manjunatha**, Shubha.M.B, Pranjal Anand, Faraaz Mohammed Ikram "Hydrothermal Synthesis and Characterisation of Hierarchically Architected Cannonite-Bi₂O(OH)₂SO₄ Nano-flowers" attend India's Biggest Nanotechnology event 'BENGALURU INDIA NANO 2018', which will be held during 5th-7th, December 2018 at The Lalit Ashok, Bengaluru
9. **C. Manjunatha**, Koushik. H. M, Abhishek. B, Mrutunjaya. Jena, Nagaraju. G, " Zn_{1-x}Al_xO nanoparticles: An excellent adsorbent for methyl orange dye derived by simple solution combustion synthesis" National Symposium on Nano Science and Technology (NSNST-2018) 20 to 22 June 2018. Indian Nanoelectronics Users Program (INUP), Centre For Nano Science And Engineering (CeNSE), Indian Institute of Science, Bangalore-560012
10. **C. Manjunatha**, Sham Aan M.P, Subana P.S "Fabrication of thiol-functionalised core shell magnetic iron oxide nano-particles for Hg²⁺ ion removal in water" under "Green Nanotechnology" Session of "World Nanotechnology Conference" which was held during April 15-17, 2019 at Dubai, UAE.
11. **C. Manjunatha**, M.B. Shubha , N. Srinivasa, S. Ashoka "Design and Development of Hierarchically Architected Highly Active polymorphs of NiS_x (x= 1, 1.33, 2) Nanostructures as Efficient Electrocatalyst towards Overall Water Splitting" in the Young Scientists' Conference (India International Science Festival-IISF-2019) to be held during November 05-08, 2019 at Biswa Bangla Convention Centre in Kolkata, INDIA.
12. **C. Manjunatha**, M.B. Shubha, N. Srinivasa, S. Ashoka, "Design and Development of Hierarchically Architected Highly Active polymorphs of NiS_x (x = 1, 1.33, 2) Nanostructures as Efficient Electrocatalyst towards Overall Water Splitting, International winter school: Frontiers on Materials science @ JNCASR, 2-6 December, 2019.
13. **C. Manjunatha**, M.B. Shubha , N. Srinivasa, S. Ashoka "Design and Development of Hierarchically Architected Highly Active polymorphs of NiS_x (x= 1, 1.33, 2) Nanostructures as Efficient Electrocatalyst towards Overall Water Splitting" in International conference of Emerging Trends in catalysis (icETC-2020) to be held during January 06-08, 2020 at VIT-Vellore, INDIA.
14. S Sarkar, V Suchithra, M Sridharan, **C Manjunatha***, "Nanobot Swarm for Targeted Elimination of Tumour in Brain" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1);SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/2959>

15. Anil Subash, C Thanmayi, GK Jayaprakash, **C Manjunatha***, "Graphene based Field-Effect Transistor biosensor: Advances in construction, working and healthcare applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/2042>
16. Anil Subash, GK Jayaprakash, **C Manjunatha***, "New hierarchical Cerium (III) Sulphate decorated reduced Graphene Oxide for electrochemical Vitamin-C sensor" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/2092>
17. PS Subana, R Suresh, **C Manjunatha***, "Remediation of Lead ions from waste water using magnetic iron oxide nanoadsorbents—a review" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/1606>
18. PS Subana, R Suresh, **C Manjunatha***, "Removal of Cadmium ions from polluted water using magnetic iron oxide nanoadsorbents—a review" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/1567>
19. BV Kumar, S Darshan, BW Shivaraj, **C Manjunatha***, "Modelling and Simulation of TiO₂/Se Sensor for Detection of CO Gas using COMSOL Multiphysics" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/2264>
20. SP Nikam, R Chandrakumar, S Ashoka, **C Manjunatha***, "Synthesis, characterization and study of synergistic effect of CuS/CuSe hybrid nanocomposite for electro catalytic application" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/2266>
21. Saad Patel, BW Shivaraj, **C Manjunatha***, "Functionalized Iron oxide nanostructures: Recent advances in the Synthesis, characterization and electrochemical sensor applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/1119>
22. AR Faisal, BW Shivaraj, **C Manjunatha***, "Advances in MoS₂ nanostructures and their composites: Synthesis characterization and hydrogen fuel generation", International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS -

23. P Hiroli, M Kumar, **C Manjunatha***, "ITO conductive ink: Advances in materials, preparation and potential sensor applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/1118>
24. A Chakraborty, S Chakraborty, M Krishna, **C Manjunatha***, "Hybrid SnO₂ nano composite: Recent advances in the synthesis, characterization and gas sensor applications", International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/482>
25. V Kumar, S Chakraborty, M Krishna, **C Manjunatha***, "Recent advances in the synthesis, characterization of pure, doped and hybrid NiO nanoparticles composite for gas sensor applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/473>
26. US Meda, YV Rajyaguru, A Pandey, **C Manjunatha**, Electrolyzers for green hydrogen generation and their integration with fuel cells, International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/946>
27. US Meda, A Pandey, YV Rajyaguru, **C Manjunatha**, "Designing a solar PV module for powering an electrolyzer" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/934>
28. 15. M Kaushik, B Vijeth, SM Darshan, GK Hegade, SK Shetty, **C Manjunatha***, "Recent advances in efficient nanostructured photocatalysts for hydrogen fuel production: A short review" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/993>
29. A Warathe, A Kaviraj, D Pradeep, S Basavaraju, Tarang Manmohan Singh Kalsy, **C Manjunatha***, "E-Nose, a versatile smart electronic nanosystem: Advances in the nanomaterials, fabrication of device, and their potential applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/992>
30. I Rahman, A Mahalingpur, S Mittal, S Shetty, R Shetty, **C Manjunatha***, "Metal oxide based electrochromic materials: Recent advances in Synthesis, characterisation and

- applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/982>
31. Sourab S, C Tarun, C Walvekar, SK Radhakrishna, LS Viranchi, **C Manjunatha***, "Advances in printable, flexible and transparent graphene photodetectors for optoelectronics applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/991>
 32. PN Amogh, HP Nandakumar, SS Vaidya, S Krishna, IR Patil, **C Manjunatha***, "Nanorobotics for advancing biomedicine: Advances in Materials, design, fabrication, opportunities and health risks" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/990>
 33. D Prabhakar, SM Abhijith, S Jain, HG Nihar, **C Manjunatha***, "Current progress in RFID Technology: Design, Materials, Fabrication and Applications" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/985>
 34. G Dheeraj, JKP Urs, SK Hombal, S Shukla, D Dwivedi, **C Manjunatha***, "Construction, working and applications of E-tongue: a versatile tool for all tastes?" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/986>
 35. NR Shekh, RP Rathod, P Kumar, MAZ Ghazi, YK Gowda, **C Manjunatha, *** "Nanomaterials embedded thin film transistor for sensor applications: Tutorial review" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/987>
 36. Hima Priya KN, Meghana C S, Neha Karunakar Raju, Shilpa S P, Yashaswini M R, **C Manjunatha***, "Current developments in conductive nano-inks for flexible and wearable electronics", International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/983>
 37. A Singaraju, VM Satyam, A Chintalapati, D Khare, **C Manjunatha***, "Advances in graphene based bioelectronic devices and applications: A concept for the future health" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/984>

38. L Saipriya, A Deekshitha, S Verma, C Swathi, **C Manjunatha***, "Advances in Graphene based MEMS and NEMS devices: Materials, Fabrication and applications", International Conference on Technologies for Smart Green Connected Societies (ICTSGS1);SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/989>
39. SA Joshi, PP Athani, PS Hegde, MS Tanmaye, **C Manjunatha***, "Potentiometric sensor devices: Current progress in materials, device fabrication and biomedical applications", International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/988>
40. S Varun, B Karthik, **C Manjunatha***, C Vidya, "Z-scheme photocatalyst for reduction CO₂ to renewable fuels: a promising and powerful strategy to curb carbon footprints" International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/902>
41. US Meda, S Deekshasushmith, **C Manjunatha**, Fire retardant nanocoatings for wooden structures, International Conference on Technologies for Smart Green Connected Societies (ICTSGS1); SGS - Engineering & Sciences 1 (01) 2021. <https://spast.org/techrep/article/view/332>.

RESEARCH PAPERS PRESENTED IN NATIONAL CONFERENCES

1. **C. Manjunatha**, D.V Sunita, H. Nagabhushana, R.P.S. Chakradhar and B.M. Nagabhushana "Combustion synthesis, structural characterization and Thermoluminescence studies on cerium doped CdSiO₃ nano powder". **National conference** on Recent Trends in Materials Chemistry and Engineering (**RTMCE – 2011, page no. 214**)" organized by Department of Chemistry, RNS Institute of Technology, Bangalore, India during September 29-30, 2011.
2. **C. Manjunatha**, H. Nagabhushana, R.P.S. Chakradhar and B.M. Nagabhushana, "Cd_{1-x}Ni_xSiO₃ nanocrystalline phosphor: Preparation, Structural characterization and Thermoluminescence properties". **National conference** on Recent Advances in Material Science (**RAMS-2011, page. no. 61**) organized by Department of Physics, M.S. Ramaiah Institute of Technology, Bangalore, India from 12–14th Dec.2011.
3. **C. Manjunatha**, H. Nagabhushana , R.P.S. Chakradhar and B.M. Nagabhushana "Synthesis, characterization and study the effect of fluxes on the microstructure,

phase, and crystallinity of CdSiO₃ nanophosphor" **National conference** on Recent Advances in Chemical and Environmental Sciences **(NCRACES-2011, page no. 84)** organized by Department of Chemistry, School of Engineering and Technology, Jain University and Prof. C.N.R. Rao Centre for Advanced Materials, Department of Physics, Tumkur University, Tumkur, India on 28th and 29th December-2011-On The Occasion Of International Year Of Chemistry-2011.

4. **C. Manjunatha**, G.K. Ramachandra, H. Nagabhushana, R.P.S. Chakradhar and B.M. Nagabhushana "Flux influence on the structural and thermoluminescence of Co²⁺ doped CdSiO₃ nanophosphor", **National conference** on Recent Advances in Functionalized materials **(RAFM-2012, page. no. 16)** organized by Department of Chemistry, M.S. Ramaiah Institute of Technology, Bangalore, India on 24-25th January 2012. (Awarded Best Oral Presentation)
5. **C. Manjunatha**, G.K. Ramachandra H. Nagabhushana, R.P.S. Chakradhar and B.M.Nagabhushana "Facile method for the preparation of microporous Cadmium metasilicate and its structural characterization", **National conference** on Recent Advances in Functionalized materials (RAFM-2012, page. no. 75) organized by Department of Chemistry, M.S. Ramaiah Institute of Technology, Bangalore, India on 24-25th January 2012.
6. **C. Manjunatha ***, Koushik. H. M, Abhishek. B, Mrutunjaya. Jena, Nagaraju. G, "Zn_{1-x}Al_xO nanoparticles: An excellent adsorbent for methyl orange dye derived by simple solution combustion synthesis" National Symposium on Nano Science and Technology (NSNST-2018) 20 to 22 June 2018. Indian Nanoelectronics Users Program (INUP), Centre For Nano Science And Engineering (CeNSE) Indian Institute of Science, Bangalore-560012.
7. G. Shireesha, **C. Manjunatha**, Anjana Jain, R. Chandramani, M C Radhakrishna, Structural and optical characterization of an inorganic nanocomposite Cd_{0.99}Ce_{0.01}SiO₃/Poly(vinylidene fluoride) flexible films, at Nanoscience-A multidisciplinary Approach, at Mount Carmel College on 10th-11th Feb 2016 at Mount Carmel College, Bangalore.

PAPERS PRESENTED IN REGIONAL COLLEGE CONFERENCES

1. G. Shireesha, **C. Manjunatha**, Anjana Jain, R. Chandramani, M C Radhakrishna, Structural and optical characterization of an inorganic nanocomposite Cd_{0.99}Ce_{0.01}SiO₃/Poly(vinylidene fluoride) flexible films, at Nanoscience-A

multidisciplinary Approach, at Mount Carmel College on 10th-11th Feb 2016 at Mount Carmel College, Bangalore.

SEMINARS OR WORKSHOPS/SEMINARS ATTENDED

2. Attended one day workshop on **"Luminescence of Nanomaterials"** on 29th January 2011, conducted by Department of Chemistry, M.S. Ramaiah Institute of Technology, Bangalore-54 in association with Luminescence of India: Karnataka Chapter [LSIKC] India.
3. Actively participated in two days workshop on **'Research methodologies and Latex'** at SJBIT, Bangalore, India conducted by VTU Belgaum, e-learning on 9th and 10th May 2011.
4. Participated in one day workshop on **"Materials for Advanced Technology"**, jointly organized by Department of Basic Science–Chemistry, Jain University, Ramanagar [Dist], Karnataka and Luminescence of India: Karnataka Chapter [LSIKC], on May 14, 2011.
5. Attended two days National Seminar on **"Recent Advances in Material Science RAMS-11"** organized by Department of Chemistry, Don Bosco Institute of Technology, Bangalore-60, India, during October 21-22, 2011.
6. Participated in one day workshop on **"Innovations in Nanoscience and Nanotechnology"** organized by Department of Physics, RNS Institute of Technology, Bangalore, India in association with Luminescence Society of India Karnataka chapter, India on August 18th, 2012.
7. Participated in the one day work shop on " Introduction to Technical Writing" organized by Dept. of Chemistry, M.S. Ramaiah Institute of Technology, Bangalore-54 in collaboration with Forum for Excellence in education, Malleshwarum, Bangalore-03 on April 7th 2012.
8. Participated in the two day on work shop " Arts and Science of Counselling" organized by Department of Placement and Training, R. V. College of Engineering, Bangalore-59, India during 10th and 11th August 2009.
9. Participated in the one day National seminar on " Research opportunities in Bio-Fuels" organized by Karnataka State Bio-fuel development Board at Visvesvaraya Technological University, Belagaavi on 19th feb 2013.
10. Participated in the one day work shop on "Inter University Accelerator Facility Workshop" organised by the Karnataka State Higher Education Council, Palace road Bangalore-01 on 27th August 2012.

11. Participated in the one day work shop on "Make in India Prospects, Possibilities and Paths" organized by (TEQIP-II Initiative) R. V. College of Engineering, Bangalore-59, on 30th January 2016.
12. Participated in the **one week work** shop on "Modelling and Simulation of Thin Film Devices" Department of Electronics and Communication, R. V. College of Engineering, (under TEQIP-II 1.2.1 initiative) Bangalore-59, India during 11th-16th July 2016.
13. Participated in the **one day work shop** on " Make in India Prospects, Possibilities and Paths" organized by (TEQIP-II Initiative) R. V. College of Engineering, Bangalore-59, on 30th January 2016.

Faculty development programme:

1. Attended the THREE DAYS "Induction programme for the Newly recruited Engineering College Teachers" organized by the Karnataka State Higher Education Council, Palace road Bangalore-01 and AMC Engineering College, Bengaluru from 21-23 April 2012.
2. Attended ONE WEEK "Faculty Development Program on Training on Advanced materials Characterisation and Analysis" organized by the Departments of Mechanical Engg., Physics, Chemistry, and Biotechnology at R. V. College of Engineering, Bangalore-59, India during 29th July to 2nd August 2013.
3. Attended TWO DAYS VTU-VGST sponsored Faculty Development Programme on "Green Chemistry and Environment" on 17th to 19th Feb. 2011 at R. V. College of Engineering, Bangalore-59, India.
4. Participated in the one week work shop on "Pedagogical skills Development" Organized by R. V. College of Engineering, (under TEQIP-II 1.2.1 initiative) Bangalore-59, India during 4th-9th July 2016.
5. Participated in the Six days Faculty Development Programme on "Recent advances in Engineering Physics" Organized by Department of Physics, R. V. College of Engineering, (under TEQIP-II 1.2.1 initiative) Bangalore-59, India during 27th Feb-4th March 2016.
6. Worked in Technical committee for the "INTERNATIONAL CONFERENCE ON SMART ENGINEERING MATERIALS-ICSEM 2016, during 20-22, OCTOBER-2016, @ R V COLLEGE OF ENGINEERING, BENGALURU.

7. Co-chair for the "INTERNATIONAL CONFERENCE ON SMART ENGINEERING MATERIALS-ICSEM 2016, during 20-22, OCTOBER-2016, Theme 2: Fabrication Techniques of Smart Devices and Structures (FTS)
Venue: Mechanical Engineering Seminar Hall

Academic enrichment programme organised:

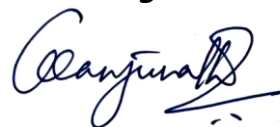
1. **Co- Chair** for the International Conference organised by "Yamagata University" on "Technologies for Smart Green Connected Societies (ICTSGS-1) held online globally on 29th and 30th November 2021. The conference attracted Over 2700 participants and published 1800 "ECS Transactions" proceedings. (<https://www.ictsgs.com/>)
2. Coordinator for One day Outreach Program on "Nanoscience and Technology: Trends and Challenges" in association with 'Centre for Nano and Soft Matter Sciences, Bangalore on 'Friday, 8th March 2019 Venue: Civil Engg. Auditorium, RVCE.

I hereby declare that the above-mentioned information is true to the best of my knowledge.

Place: Bengaluru

Date : 07-05-2026

Signature



(Dr. MANJUNATHA. C)